



GE Medical Systems

Technical Publication

Direction 2243321-100

Revision 10

LightSpeed 2.X

Application Tips and Work-Arounds

Copyrighted © by GE Medical Systems, 2003



Page Left Intentionally Blank

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call Traffic and Transportation, Milwaukee, WI (262) 785 5052 or 8*323 5052 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage. Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins. 14 July 1993.

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment. Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives. GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Page Intentionally Left Blank

LEGAL NOTES

COPYRIGHTS

All Material Copyrighted(c) 2003 by the General Electric Company, All rights reserved.

REVISION HISTORY

Revision	Date	Reason for change
0	10/13/00	9.3_2.8.2I_H2M3_H1.3M4 Pilot Release
1	12/8/00	11.3_2.8.2J_H2M3.1 Pilot Release
2	3/22/01	12.2_2.8.2K_H2M4 Release
3	5/3/01	13.16_2.8.2L_H2M4.1 Release
4	8/7/01	14.11_2.8.2L_H2.1M4 Pilot Release
5	8/24/01	14.13_2.8.2L_H2.1M4 Release
6	10/30/01	14.13_2.8L_H2.1M4 + patch ver1.sw.mobile Release
7	11/5/01	2.2m3.11c_2.9E_1.1_20011011_H2.2M4 Release
8	6/20/02	2.3.3_H2.3M3+Patch
9	10/31/2002	2.3.7_H2.3M4 Release
10	11/14/2003	2.4.2_H2.4M5 + patch

Page Left Intentionally Blank

TABLE OF CONTENTS

Legal Notes.....	5
Table of Contents	7
New Features Version 2.4.2_H2.4M5 Software.....	10
SOFTWARE VERSION	10
CHANGES TO SHUTDOWN	10
REPEAT SERIES	10
AUTO MA	11
PROVIEW	11
LOW SIGNAL CORRECTION	12
AUTO TRANSFER.....	12
ANONYMOUS SCAN DATA SAVE	12
IMAGE ANNOTATION	13
CUSTOM ANNOTATION.....	13
New Precautions 2.4.2_H2.4M5	15
SYSTEM.....	15
SCAN.....	15
TUBE WARM UP/FAST CAL	16
PERFORMED PROCEDURE STEP (PART OF CONNECT PRO OPTION)	17
PROTOCOL MANAGEMENT	17
SMARTPREP (PURCHASED OPTION).....	17
RECONSTRUCTION	17
RECON MANAGEMENT	18
DISPLAY (APPLIES TO BOTH IMAGE WORKS AND EXAM RX DISPLAY).....	18
IMAGE WORKS DISPLAY	18
EXAM RX DISPLAY.....	18
VARIVIEWER	19
DIRECT3D - OPTION	19
REFORMAT.....	19
CT PERFUSION 2 ON OC – PURCHASED OPTION	20
ADD SUBTRACT	20
ARCHIVE	20
FILMING	20
NETWORK.....	21
IMAGE MANAGEMENT	21
LEARNING AND REFERENCE GUIDE	21

Legacy Precautions	23
SYSTEM	23
TUBE WARM UP/FAST CAL	24
SCAN	25
PROTOCOL MANAGEMENT	27
SMARTPREP – PURCHASED OPTION	28
PATIENT SCHEDULE	28
CONNECTPRO (PURCHASED OPTION)	28
PERFORMED PROCEDURE STEP (PART OF CONNECTPRO OPTION)	29
RECONSTRUCTION	29
RETRO RECON	30
RECON MANAGEMENT	30
VARIVIEWER	31
SMARTSTEP – PURCHASED OPTION	32
DIRECT3D – PURCHASED OPTION	32
CT PERFUSION 1 ON OC – PURCHASED OPTION	33
CT PERFUSION 2 ON OC – PURCHASED OPTION	33
SMARTSCORE PRO – PURCHASED OPTION	33
CARDIAC SNAPSHOT – PURCHASED OPTION	33
AUTO TRANSFER	34
DISPLAY	34
EXAM RX DISPLAY	35
IMAGE WORKS DISPLAY	36
EDIT PATIENT DATA	37
3D/NAVIGATOR/DENTAScan/ADD/SUBTRACT	37
ADD/SUBTRACT	37
REFORMAT	37
FILMING	38
ARCHIVE	39
NETWORK	40
IMAGE MANAGEMENT	41
MOBILE	42

Page Left Intentionally Blank

NEW FEATURES VERSION 2.4.2_H2.4M5 SOFTWARE

SOFTWARE VERSION

LS2002M5 – This release of software contains fixes to address a number of issues that were reported on the LS2002M4 release concerning reconstruction retirees for Scout images, intermittent loss of alignment lights and prescribed tilt capabilities, Show Localizer not displaying or having a mismatch, ScanRx shutdowns, AutomA changes to improve reliability, image quality fixes for ring artifact with Bone and Bone Plus algorithm, improved availability of images in the AutoLink viewport.

There are some new features as well as enhancement to existing features.

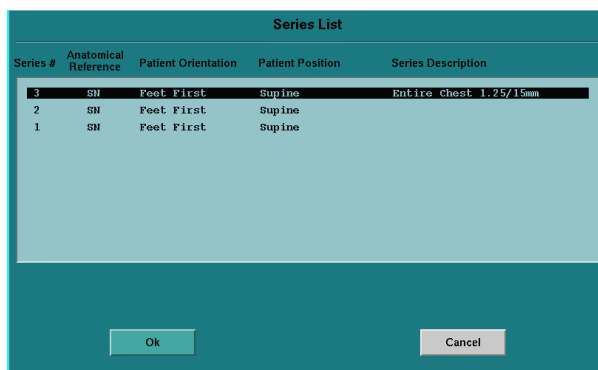
CHANGES TO SHUTDOWN

Shutdown - When shutdown is selected, you will be presented with 2 choices Restart or Shutdown. Restart shuts the system down and then automatically restarts the system. You no longer need to “Press any key to restart” the system. Shutdown brings the system down to the power off prompt. When shutdown is selected, you can to cycle power to the operator console to reboot the system.



REPEAT SERIES

Repeat Series - Additional functionality for Repeat Series has been added for this release. Repeat Series now allows you to choose any series that has been scanned. When more than 1 series has been scanned a list of all scanned series will be displayed. Click on the series that you wish to repeat. If only one series has been scanned, Repeat Series will not display the Repeat Series menu.



AUTO MA

AutomA - changes have been made to the user interface to simplify the selection of parameters.

The **mA Control** dialog box has a teal background. At the top, it is titled "mA Control". Below the title bar, there are two main sections. The top section is for "Auto mA" and includes a "Reference Noise Index" field with the value "11.57", a "Reset" button, a "Dose Steps" field with a double-headed arrow and the value "+0.00", and a "Noise Index" field with the value "11.57". Below this, there is an "mA Range" section with "Min" and "Max" fields, showing values "10" and "440" respectively. The bottom section is for "Manual mA" and includes a field with the value "120". At the very bottom, there are "OK" and "Cancel" buttons.

Reference Noise Index	Default or baseline Noise Index for the given protocol. Any changes to Dose Steps, Slice Thickness or Noise index are referenced to this value. This value can only be defined while in Protocol Management.
Noise Index	The noise level required for the study. As the Noise index increases the required mA decreases and image noise increases.
Dose Steps	Adjusts Noise Index by steps of 5%. Dose steps can be increased or decreased + values decrease image noise thus increasing required mA. Minus values increase Noise index, thus decreasing required mA. A Dose step value of 0 indicates that the prescribed Noise index is equal to the Reference Noise index for the protocol.
Reset	Resets the Reference Noise Index to the GE Target Noise Index Default for the anatomical area and slice thickness chosen in the protocol

PROVIEW

ProView Filters – the number of filters have been expanded. For Edge, the filters E21, E22, E23 have been added for finer control for edge enhancement factors. For Smooth, the filters S11, S21 have been added for finer control for smoothing factors.

The **Select Image Filter** dialog box has a teal background. It is titled "Select Image Filter:". Below the title bar, there are two rows of filter buttons. The first row is labeled "Edge" and contains buttons for "E1", "E2", "E21", "E22", "E23", "E3", and "Lung". The second row is labeled "Smooth" and contains buttons for "S1", "S11", "S2", "S21", "S3", and "Off". At the bottom, there is a "Cancel" button.

LOW SIGNAL CORRECTION

Advanced Artifact Reduction – low signal reconstruction correction has been updated for this release. A reduction in streak artifacts through shoulders and hips should be seen.

AUTO TRANSFER

Auto Transfer - by image has been enhanced to transfer images in larger groups. This provides more efficient use of system resources while Auto Transferring by image. Auto transfer by image will transfer the first 50 images in 2 groups of 25, after that images will be transferred in groups of 50.

ANONYMOUS SCAN DATA SAVE

Anonymous Scan Data Save - The system now provides the ability to save scan data with patient identifying information anonymized. This has been added to address HIPAA concerns about patient confidentiality. Scan files (raw scan data) saved anonymously and then restored will reconstruct with patient identifying information removed. The Patient Name and ID will be anonymized.

Select Recon Management

Select Save Scan Data

Select Save Selected Anon. Data to save scan files as anonymous data.

Recon Management Save Scanfiles

Examinations					Scans						
Suite	Exam #	# of Series	Date/Time		Suite	Exam #	Series #	Scan #	Type	# of Views	Date/Time
B20A	414	1	5/16/2001 12:32		B20A	414	/4	/1	Axial	992	5/16/2001 12:32

Series											
Suite	Exam #	Series #	# of Scans	Date/Time							
B20A	414	4	1	5/16/2001 12:32							

Examinations					Scans						
Suite	Exam #	# of Series	Date/Time		Suite	Exam #	Series #	Scan #	Type	# of Views	Date/Time
B20A	414	1	5/16/2001 12:32		B20A	414	/4	/1	Axial	992	5/16/2001 12:32

Series											
Suite	Exam #	Series #	# of Scans	Date/Time							
B20A	414	4	1	5/16/2001 12:32							

Buttons: Save All Scanfiles, Deselect All Exams Listed, Select All Exams Listed, Prior, Next

Buttons: Save Selected Scanfiles, Save Selected Anon. Data, Deselect All Scanfiles Listed, Select All Scanfiles Listed, Prior, Next

IMAGE ANNOTATION

Accession Number – Image annotation field Req No.: has been replaced with annotation which more appropriately identifies it as the field for the accession number entered on the New Patient screen. The annotation on the image is Acc No.:

CUSTOM ANNOTATION

Custom Annotation – has added an additional custom annotation field for Accession Number. Selecting group 17 will allow the user to enable or disable display of the Accession Number field on the monitor or on film.



NEW PRECAUTIONS 2.4.2_H2.4M5

SYSTEM

- The scan hardware may fail to reset on reboot. Select Service Desktop, System resets, Scan, Run to reset the scan hardware. If the reset still fails, then shutdown and reboot the system to recover.
- The system may not show typed information on the New Patient or ViewEdit screen. The system will update after a short time.
- The system may be slow to respond to keyboard entry and mouse clicks. If the system fails to respond stop and let the events catch up.
- The first character in the Exam description may not be shown in the text field but is shown in the Browser description.

SCAN

- Show localizer does not show the Scan Field of View when prescribing scans. When scanning head studies, verify that you have your patient centered at ISO center to assure that the entire head is included in the SFOV.
- Show Localizer may fail to display. If toggling the Show Localizer button does not display the scout, then display the scout in a free viewport and use Report Cursor to find your scan locations and A-P and R-L centers, manually enter these on the ViewEdit screen.
- Show Localizer may remove the localizer lines when page down or page up is selected and will not advance to the next scout. To recover from this, toggle Show Localizer off /on. If Scout Image 2 is displayed use Page Up, to go to scout image 1 use Page Down.
- Show Localizer may display a scout from a previous patient along with the current exam scout. Even though this scout is displayed, the scan prescription lines will not be shown on this scout.
- The Show Localizer button may get in a stuck state and not display a scout image if the last scout scanned has an azimuth other than 0, 90, 180, 270. Manually display a scout in a free viewport and use report cursor to find your scan locations and then manually type in the Start and End location, RL and AP Centers as desired.
- A scan abort may occur between 2 scan groups with gating enabled. To avoid this prescribe the shortest delay possible.
- Auto Voice may fail to play for prescriptions with 2 groups and Auto mA enabled or with manual mA when the mA changes between the groups. An Auto Voice has failed to play message will be posted on the right monitor in the message area on the upper left of the right screen.
- The pause button on the screen in Tube warmup does not function. Use the stop scan button on the SCIM.
- You may be prompted to run DAS gain cal prior to starting Fast Cal.
- The system may not allow changing of the landmark even though no scans have been taken in the series. A landmark that is set at the edge of the scannable range most often causes this. To reset the landmark, choose Select New Protocol.
- Always verify the mA table prior to scanning when using Auto mA.
- The dynaplan screen may fail to update is selected during scanning.

Scan continued

- Auto mA may set the min and the max value to 10 when selecting a protocol or adding a group. Always verify the mA tables prior to confirming the scan to verify the mA values in the table are reasonable for the patient being scanned. If the values do not look reasonable select the mA button and verify the Auto mA parameters.
- Auto mA may show the max mA from a previous group instead of the max mA prescribed. Always verify that the max mA parameters are correct before confirming a scan.
- Auto mA may set the Max mA to 50. Always verify the mA table prior to scanning.
- The gating button may turn red when the patient experiences a big jump in heart rate such as in a PVC. The system usually will recover and read the signal. Sometimes you may have to toggle the gating button Off/On to get the heart rate signal synched again. Care should be taken in starting the scan if the patient continues to have PVC's
- The cut lines in Show Localizer and in Cross Reference in the vertical presentation appear multi-colored on the monitor. The lines are a solid color when printed on film or screen saved.
- Heart rates of greater than 120 are not valid for prospective gating for SmartScore acquisitions. The system will not stop scanning if the heart rate exceeds 120 BPM; however if the user observes higher than acceptable heart rates, they should pause the scan and resume when heart rate is in a valid range.
- The Start/End locations and Gantry Tilt fields may show strange characters when [Add Group] is selected quickly after [Show Localizer] is selected and a Direct Vis application (Direct 3D, VariViewer) is enabled for the series. Wait until the localizer is displayed before performing any edits for the series.
- Scan Abort may occur if firmware detects the Axial drive not ready. Select Resume and continue.
- AutoVoice may fail to play if the table is moved between groups where a long group delay has been prescribed.
- Alignment lights may fail to turn on. Perform a system reset. If alignments lights still fail to turn on. Shut the system down and turn off power at the main breaker and then restart system.
- Show Localizer may remove the slice lines when page down or page up is selected and may not advance to the next scout image. To recover, toggle Show Localizer off / on.
- System performance may become slow if the optimize screen for mA is left open. Make selection in the optimize pop-up screen and then close the window by clicking on Optimize button.
- Selection of up front delay in Optimize is very slow, it may take 10-20 seconds for the system to respond.
- Add group may not work. Try selecting One More and changing the number of slices to desired number.
- In GraphicRx, it maybe difficult to select a group that is inside the range of another group. Select group on the ViewEdit screen to make that group active in Show Localizer.
- If you have two protocols within the same exam that have different patient positions, the system may not move to the correct start location for the series with changed patient position. If patient position is going be changed in regards to head first verses feet first, please start a new exam.

TUBE WARM UP/FAST CAL

- Fast Cal may give error that Collimator Cal failed. Resume Fast Cal and contact GE Service.

PERFORMED PROCEDURE STEP (PART OF CONNECT PRO OPTION)

- The PPS server may stop and fail to send the request. You may have to select Complete multiple times to get the series to update to the complete state.
- Add Sub does not update images for PPS correctly
- Perfusion2 will not launch if images contain PPS information.
- If the total number of prospective images in an exam is greater than 2500, PPS may fail to complete.

PROTOCOL MANAGEMENT

- The Dose information in protocols used from Most Recent does not reflect the dose actually used for the protocol. If the protocol contains manual mA, retype the mA value to update the dose display. If Auto mA is used, the dose will not be reflected in Protocol Management, but will be updated at scan time for the patient being scanned based off a valid scout image.

SMARTPREP (PURCHASED OPTION)

- SmartPrep may fail to confirm to take the baseline image. Try confirming again, if this does not clear the problem then a system reboot will be needed to get SmartPrep working again.
- The SmartPrep baseline image may fail to display even though it is recon'd and in the database. Go back to ViewEdit and rescan the baseline image to continue. The Monitor phase image and graph information will not update if the table height has changed from the Baseline to Monitor phase. Do not change the table height after the baseline scan has been taken to avoid this.
- When transitioning to the Scan Phase from Monitor phase if the XR Auto View layout is selected the upper left viewport will be black. Move your mouse cursor over the viewport to redisplay the image.

RECONSTRUCTION

- Streak artifacts may be seen in images when a tube spit has occurred.
- Image recon may fail and then shutdown. Select Recon Management, select Pause queue, Restart queue, if this does not restart recon then a Restart of the system will be needed.
- Cardiac recon modes of Burst 2 or Burst 4 cannot be selected via column edit. Please select the individual row to edit recon mode selection.
- The dynaplan screen may fail to update if Priority Recon is selected during scanning
- Anatomy scanned off center from ISO center such as extremities, may have loss of resolution when data is acquired helically. Always make sure to center to anatomy of interest as close to ISO center as possible.
- Cine acquisitions may have retries or image suspension. Unsuspend images in recon management.

RECON MANAGEMENT

- Scan File save is not complete even though the Saved File message has been posted. The problem is due to the time it is taking to unmount the MOD. You may have to wait 5-7 minutes for the MOD to unmount. The write light on the MOD will be green when the MOD is unmounted
- Scan Files restored from a different system will be reconstructed using the product name of the system it is restored on. For example, restore scan files from a H1 SDAS QX/i or a H2 SDAS Plus to a LightSpeed¹⁶ and the product name on the images will be indicated as LightSpeed¹⁶ instead of LightSpeed QX/I or LightSpeed Plus. All other patient and parameter information is correct.

DISPLAY (APPLIES TO BOTH IMAGE WORKS AND EXAM RX DISPLAY)

- An active user annotation graphic will not film the box or arrow that is displayed on the screen.
- Font size for the ROI Text Page is smaller compared to prior release of software.

IMAGE WORKS DISPLAY

- The primary focus may change if the space bar is selected multiple times when entering accelerator line commands. Verify that primary focus is on the image desired.
- The Accelerator Line is a dark green in Image Works instead of a light green as on the ExamRx desktop. It is active and will accept commands.
- When changing from 3D to Navigator, sometimes the threshold value is automatically set by the system and the user is unable to change it.

EXAM RX DISPLAY

- The Trackball may fail to initialize properly at system reboot. Try rebooting the system again to recover
- The Icon for paging/WL and Auto Link is not visible
- Display windows may display on the left screen instead of the right. This will most often occur when switching desktops rapidly. A system reboot will be needed to recover.
- Pressing page up 2 times causes the ww and wl to change on the scout on it's own. Readjust the ww and wl to continue.
- The first time the trackball is used for paging after a system reboot window level will be changed instead of paging the images even though the P icon is displayed. Click once to switch to W mode then click again to place the trackball in paging mode.
- The AutoView or Auto Film viewport may display a very magnified image. This image may be filmed in Auto Film. Review your films and verify that all filmed images look correct.
- Viewport format may not be able to be changed and requires switching layouts to correct. This occurs if F3 or Shift F3 is used for filming and next image is used. The viewport format is now locked. You will have to switch to another review or Auto View layout to correct.
- MIROI may fail to produce a graph if next/prior is selected while creating the ROIs. Redisplay the series and begin again.
- Cross Reference image may not print sometimes instead the last image will be printed twice.

VARIVIEWER

- Batch film may fail after a system reboot if the format selected is different than the default film format on reboot. An error dialog will be posted informing you of the error. You may have to select filming multiple times to successfully send the request.
- If the scout is taken after the series used for VariViewer operation, the system may be unable to find the scout for graphic batch prescription. Manually enter the range for the batch prescription.
- Any zoom applied to a Variviewer image is not maintained as you page through the images.
- VariViewer session may not be able to be started if there are multiple Pause scans during the acquisition of data.
- VariViewer is not compatible with a gantry tilt.
- Switching back from a 768 viewport is VariViewer causes the VariViewer viewport to be forward instead of the Auto View viewport. Use the page turner to toggle back to the auto view Viewport.

DIRECT3D - OPTION

- Batch film may fail after a system reboot if the format selected is different than the default film format on reboot. An error dialog will be posted informing you of the error. You may have to select filming multiple times to successfully send the request.
- 768 image size may display split in 2 if resize is selected with the Auto View viewport forward. Always have the Direct3D viewport forward when selecting resize. To recover from this problem select resize twice.
- If you resize the viewport in Direct3D, the image may not initially be displayed. User the Page Turner in the lower left corner of the viewport to toggle to the time.
- The Direct3D model may not initially build properly. Enter Interactive Review and re-render the model and then the model will build correctly.
- The Save State, which allows the user to rebuild the Direct3D model on the AW using the VR algorithm from D3D, does not work on the OC even if Volume Viewer is installed. The user will get an error message informing that the images are missing.

REFORMAT

- Reformat will only allow filming in Batch to the first 30 cameras in the list. If another camera is desired save the batch images to disk and use Print Series to film the images.
- Images saved from batch mode may appear to have jagged edges or not have optimal Image Quality. Manually save the images to improve image quality.
- Large image series may fail to load when using Reformat detail. This can be caused by small spacing interval between images or image sets greater than 1000 images. Select a sub set of images if this occurs or use Reformat Standard to build the model.
- A series that has more than 1 group with 0.625mm slice thickness will fail to load for reformat or will fail in Volume rendering on the AW. This is due to a mismatch in the spacing between the groups. To avoid this have only one scan group if possible, otherwise selectively highlight the images from the first group for Volume Analysis.

Reformat continued

- Sagittal images created from a head first prone data set scanned in the direct coronal position will appear elongated.

CT PERFUSION 2 ON OC – PURCHASED OPTION

- Computation of perfusion maps and automatic detection of the artery or vein may not function when the data displayed in Perfusion is initially black and the WW and WL values need to be set to large negative values (WW 40, WL –1020) to make the images viewable. The threshold should be set to the maximum negative value for air to assure that automatic detection of the artery and vein and that Perfusion maps are displayed correctly.
- Perfusion will not launch if the images contain PPS information.
- Saving TIFF files of the graph and images is not supported because there is no floppy drive hardware available on the OC.
- Set the color ramp to gray scale for screen saves images if you are going to network the images to another station such as PACS to improve quality of the screen save.
- Initially as CT Perfusion2 is loading, you may see garbage flash along the left side of the screen.
- Processed data does not display if New Protocol is selected within Perfusion2. To avoid this always close Perfusion2 before selecting a new protocol.

ADD SUBTRACT

- Comb images will not contain the e/s/i used for the added image.

ARCHIVE

- Even though the Sony MOD drive states that it is a 5.3 gb drive. The archive software only supports a maximum size of 2.3 gb. 2.3 gb 512 bytes per sector is the preferred archive media size to utilize on the system.
- Archive may fail and slow the system when more than 200 images are queued by image. Try to archive by series if possible to avoid this.
- The system may report that the media is full even though the media has just been labeled when Save by image is selected. Try selecting a smaller range of images to avoid this.

FILMING

- Image setting such as WW WL and flip rotate, zoom, roam are not maintained across all images when imitating F4 print series from a MID Viewport format. Use a 512 size viewport to initiate F4 Print Series.

- Auto Film may stop and the Continue Same Sheet button will be displayed. The Continue Same Sheet button will be active, but when selected will not restart filming. Use Manual Film to complete filming the exam and reboot the system before starting the next exam
- If the F1 key is used to film the Text Page ROI, Exam Text Page and Series Text Page, the text pages will be filmed to the film composer last selected in the pop-up for text page. Use the filming selections in the text page screens to film the information displayed.
- The message "Film formatting in progress. Please retry." May be seen when filming to the Manual Film composer when the system is busy with other simultaneous operations such as recon, network, auto film and scanning. Retry manual filming when the system is less busy. A system reboot may be required to recover.
- Image annotation may be small or blurry on filmed images when the first image is a scout that is over 600mm. Place this image on the last space of a film to avoid this.

NETWORK

- The system may create core files that may fill up disk space on the system. Networking color save images from Volume Viewer or CardiacIQ causes these core files. To avoid this save the color images to MOD, remove the color images from the system then restore the color images and then network the images.
- Color Saved images from Volume Viewer or CardiacIQ may fail to transfer. To avoid this save the color images to MOD, remove the color images from the system then restore the color images and then network the images.
- Images that have been edited using Edit Patient and then networked to a RadStore system will fail to archive to the RadStore DVD. The images must be archived on the CT system MOD.
- The system no longer supports Advantage Net protocol. When sending images DICOM send should be utilized.

IMAGE MANAGEMENT

- Removal of images should be done with the system in an idle state. Failing to remove images at idle times can cause image install timeouts that can then cause recon to shutdown and can cause the browser to fail to respond. The user may also see a blank list in the browser. The system should not be shutdown until the image space has been updated fully from the remove. The length of time to synchronize the database is proportional to the number of images being removed. This will assure that all images are reconstructed, displayed and installed into the database without error and that image space is update correctly.

LEARNING AND REFERENCE GUIDE

- After a desktop switch back to Learning Solutions desktop the guide returns to the home page instead of returning to the page they were on when the switch occurred. The document is slightly behind the home page. Left click and drag on the home page border to move the home page so you may access the Learning and Reference Guide pdf file.
- The Learning Solutions CD-Rom may not launch the first time. Wait a minute and try again.

Page Left Intentionally Blank

LEGACY PRECAUTIONS

SYSTEM

- Run Shutdown and Restart the system daily.
- The system may fail to start if system disk space needs to be recovered from unused system files. The following message will be posted in the Disk Management shell window:

```
"The purpose of storelog is to recover system disk space by archiving and then removing core, log, and data files that have been saved for their system troubleshooting diagnostic value. Now they may be taking up too much space for the system to run properly. Removing these "system log" files does not add image space, but should allow the application to start."
```
- You are then directed to answer the following questions (be patient for the messages to display):
Place the mouse cursor inside the black Disk Management window
Q: Do you want to save system log files to removable media
Type n and hit the enter key
Q: Do you want to remove the system log files
Type Y and hit the enter key.
You will then see a message "Please be patient, the system is now removing unwanted files." The system will then start normally.
- If the system fails to startup completely, select [Unix Shell] from the tool chest menu in the upper right of the Scan Monitor place your cursor in the blue shell and type st.
- If mouse selections fail, select the [Escape] key to clear backlogged requests.
- If you can no longer type in a field, move the cursor to the field and try clicking the middle mouse to restore functionality. If this does not work, then a system shutdown will be required.
- The cursor may not move between the left and right screen. Wait a few seconds without moving the cursor and try again.
- To install a SMPTE pattern, BRH or Quality assurance images for viewing, select the following. Once installed the images will be available for viewing from the Image Works browser or List Select in Exam RX. Both will be listed as Exam 1000, the patient name will reflect if it is a QA image or a SMPTE pattern.
 - Select Service Desktop
 - Select [Diagnostics]
 - Select [Display Processing]
 - Select [Install SMPTE Image]
- In general, wait for a screen transition to take place before making another selection.
- If the console becomes unresponsive for 2 minutes or more, shutdown the system using the pink shutdown button and restart the system. If you cannot select the pink shutdown button, turn off the console power switch, wait 10 seconds, then turn the console power switch back on. The system should come up normally.

System continued

- The system will lock up if the mouse becomes hung on the left monitor. The only known recovery is to cycle power to the console.
- You may see a pink warning dialog posted indicating the system has low disk space. This is due to a partition on the system disk getting too full. This can be caused by archive, network or film queues containing too many enteries. Make sure that the enteries are processed. If the queue is paused, resume it or delete enteries as needed.
- Protocol Management may not be selectable. Check that retro recon is selectable. If it is not then the system will need to be rebooted.
- Pop-up screens and menus may appear on the wrong monitor or may be displayed split between the two monitors.
- Windows such as the Film Composer which normally are restrained to the right or left monitor can be dragged to the other monitor. Be careful not to get a Film Composer hidden behind some screen if you move it to the other monitor.
- If a blank viewport is seen where image annotation is displayed and W/L is interactive but no image is seen, shutdown the system, recycle power and reboot the system to clear this display issue.
- If corrupted images are seen after applying an image filter such as enhance, smooth, lung or gray scale, try rebooting the system and see if the image can be displayed correctly. If not contact your GE service representative.
- Pressing the Space Bar after selecting any of the fields in New Patient locks the patient information fields. All the patient information fields become insensitive. Only the delete and backspace key will be active. Press the delete key to recover from this situation.
- Entering a \ and pressing the Enter key in the New Patient can cause the fields to become insensitive.
- Double and triple click for selection of primary or secondary focus may be slow. Wait for the system to respond before repeating the focus selection mouse commands.
- Auto Voice volume setting changes are not remembered across system reboots unless the change has been saved. The save a volume setting do the following:
 1. Select Image Works
 2. Select Auto Voice Volume
 3. Adjust the settings as desired
 4. Select File menu
 5. Select Save
 6. Select OK

TUBE WARM UP/FAST CAL

- The detector temperature has a 15 minute stabilization time immediately after a system restart. If Fast Cal is entered during this stabilization period a dialog will be posted instructing the user to wait x minutes before beginning Fast Cal.

Either wait or run Fast Cal later. Do not forget to run Fast Cal daily.
- To avoid having to wait to run Fast Cal, do one of the following:
 - Run Fast Cal before performing Shutdown/Startup
 - Perform Startup/Shutdown and Fast Cal separately

Tube Warm Up/Fast Cal continued

- Run Tube Warm up after 2 hours of non-use. It is best to warm the tube as close as possible to the time the next patient will be scanned. Don't warm the x-ray tube and then wait 30 minutes without scanning.
- Always check that the beam is clear when doing Fast Cal. In Fast Cal, the system checks cleanliness of the Mylar Window. If the system suspects the Mylar Window may be dirty to the point that it may cause a beam obstruction, a pop-up message will be displayed. The operator will be instructed to clean the Mylar Window and select [Retry].
- Fast Cal must be run once every 24 hours.
- Complete all portions of Fast Cal, Warm Up 1, Warm Up 2, Gen Cal, Clever Gain and Fast Cal. This assures that the Air calibration and generator calibrations are up to date on the system.
- A warning message will be posted if detailed calibration is selected with more than 2 kV stations. To run detailed cal on more than 2 kV stations requires covers at the rear bottom of the gantry to be removed. Please work with your Field Engineer on detailed calibrations issues.

SCAN

- Scan aborts may occur during Axial or Helical scanning. Always be aware of the scan progress during an Exam and select Resume as soon as it is posted to continue.
- To terminate an Insite connection on your system. Select [New Patient] a message will be posted informing you that the scan hardware resource is not available. Wait 3 minutes and select [New Patient] again to begin scanning.
- In general, if a scan fails and a [Resume] is posted, press [Resume] to continue. Try [Resume] again if the first [Resume] fails. If a failure still occurs, reset the scanning hardware through System Resets in the Service desktop. If scan still fails to restart, shutdown and Restart the system.
- Auto Voice may fail to function, especially during system simultaneity. Make sure that you can hear the Auto Voice to recognize if Auto Voice has quit, manually breathe the patient when this occurs.
- The Show Localizer scout image may fail to display or images may fail to recon and may become suspended if scanning is started while remove images is in progress. To avoid this **don't** remove images while an exam is in progress or scanning is active. Remove images when the system is idle.
- The Dynaplan screen may report incorrect status (i.e. a scan is removed before it is actually scanned) when stop scan is selected. The screen will reset correctly after Resume or back to view edit is selected.
- Scan groups will not be contiguous if you switch from 2 or more Helical scan groups column or row edit and the scan type is changed to Axial. If this change is made, then verify the Start and End locations of each of your scan groups and adjust if needed.
- The Cine time between images may change when other parameters are modified within a protocol. Check the Cine time between images prior to confirming the scan to verify it is the value you wish.
- Tilt handles will not be visible on the screen if the DFOV is larger than 48. Use a DFOV smaller than 48 to assure that the tilt handles are visible.
- Show Localizer may fail to display the scout image if the Next Series is selected before the scout images are reconstructed. Wait until the scout images are reconstructed before selecting Next Series. If Next Series has already been selected, toggle Show Localizer off and then on again.

Scan cont.

- Quickly changing desktops when Show Localizer has been selected for the next series can cause a partial display of the scout in the Graphic Rx window to occur.
- If Page Up or Page Down fails to change the displayed Show Localizer.
Try the following in order:
 1. Turn off then on the Show Localizer button
 2. Move the cursor off the image window and back on to the image to refresh the screenIf both of these fail then verify that you have a valid scout for Show Localizer.
- When changing from a 3.75mm/11.25 HQ scan to HS mode, the slice thickness will change to 5mm/22.5 with no notice to the operator. The 3.75 thickness is not valid at 22.5 table speed. If 3.75mm thickness is desired, then change your table speed.
- To copy Start/End location, Number of Images, Thickness, Interval, DFOV, R-L center and A-P center from Recon 1 to Recon 2 or 3 toggle Recon 2 or 3 from N to Y. This will need to be checked anytime you change the Start and End for Recon 1 when Recon 2 or 3 is turned on.
- If a R, L or A, P value is more than half of the Display Field of View in mm. The image annotation will not annotate R and L or A and P on the images. For example if the DFOV is 10cm and the R value is 56mm then the image annotation will show R R. Use a R-L or A-P value which is less than half of the DFOV in mm to avoid this.
- A protocol may fail to display. The following message will be posted to the message line on the lower center of the scan monitor "Can't read selected protocol, please choose another protocol". If this occurs the protocol is corrupt and will need to be rebuilt. Delete the protocol and rebuild it.
- The patient history field may be missing the last 9 characters when all 60 characters have been entered and the patient record is either selected as a completed exam or the Patient ID is entered and the system matches a completed exam.
- Even weights entered in Patient Schedule may be rounded up to the next odd pound. This occurs because weights are stored as kilogram units then converted back to pounds.
- During the acquisition of large data sets (1000-1500), there may be Autofilm and reconstruction performance issues. If either stops, check the queues and restart to continue.
- When selecting the Auto Voice field, the selected Auto Voice will be deselected. You must reselect the desired Auto Voice message before exiting the Auto Voice screen.
- When entering birthdates on the New Patient or Schedule Patient screens the following rules apply:
 - 2 digit years can only be entered if the birth year is in the current century for the 21st century - for the year 2000 forward.
 - Birthdates from any other century must be entered as 4 digits for example for the year 1899 – 1999 all four digits must be entered.
 - Birthdates can only be entered for 120 years (Current year minus 120 years).
- If InSite is running the "Remote Safety" test, New Patient will not open. New Patient can be made available by either:
 1. Calling InSite and request they abort the test.
 2. Going to the Service Desktop and selecting [CleanUp] option to cancel the test in progress.
- PMR images may fail to recon if disk space is low. To recon the images, remove images and restart the recon queue in Recon Management.

Scan cont.

- You may get the message “Duplicate Scan Key”, if a Scout Scan is aborted within the first 20 to 75 mm of table travel or if a the prescribed length of the Scout is less than 75mm. In order to proceed, an End Exam must be done and new exam started.
- If a large number of scan files are reserved and there is a large recon queue, you may see the message: “Duplicate Scan Key – unable to allocate scan file.” There will be a resume at Start Scan. Either wait a few minutes for the recon queue to decrease or release some of the reserved scan files.
- Bands and lines may be seen in scout images if there is a tube spit or the reference channel is blocked during the acquisition of a scout scan.
- A zero interval axial series, if paused, will add another group equal to the remaining number of images. Make sure to return to the View/Edit screen to delete this group to avoid acquiring additional images.
- If the message “Can not read Cal Database”, select [End Exam] and re-enter patient information.
- Unable to confirm to scan because not enough scan keys. The following message will be seen in the gesyslog: “MA non-zero errno was encountered during a remote system call. The system cal may have failed to complete successfully or may have been interrupted.” Shut the system down and cycle power to the system two times.
- For Thin Twin (sub mm) acquisitions, only 0.625 images can be made even though the interface show 1.25mm available.
- If the scout is not displayed but available in the Browser and you get the message “Patient Position has changed” or “No matching Scout found” for show localizer. The system will not be able to perform graphic prescription. Use explicit prescription to complete the prescription or end the exam and start a new exam.
- If Direct3D or VariViewer are enabled in the DirectVIS under the Recon tab, Add Group will display some fields as insensitive as it will be combined with the current Direct3D or VariViewer session. Turn Direct3D or VariViewer off for the added group if you no longer want it to be included as part of the Direct3D or VariViewer session. This will allow changes to any of the acquisition parameters.
- Auto mA should not be used with Gating and Cine or Cardiac Helical acquisitions even though it can be selected.
- The gating button may turn red when the patient experiences a big jump in heart rate such as in a PVC. The system usually will recover and read the signal. Sometimes you may have to toggle the gating button Off/On to get the heart rate signal sync'd again. Care should be taken in starting the scan if the patient continues to have PVC's.
- Hitting the F12 key while in Show Localizer can cause ScanRx to crash. Once the system is in the Show Localizer mode to perform graphic prescription, the F12 key has no functionality. F12 key is valid only in AutoReview and Review Layout formats.
- Auto mA may default to a Noise Index of 10. Change the slice thickness and return to force the system to read the default noise table.

PROTOCOL MANAGEMENT

- Show Localizer ON is not saved for protocols copied or used from Most Recent tab even though it was turned ON for the protocol. Edit the protocol after it is copied or selected to turn Show Localizer ON.
- Protocols with SmartStep series in them cannot be saved as a protocol from Most Recent tab. SmartStep cannot be prescribed prospectively in a protocol.
- If there are pre-existing pediatric protocols in the body categories, they will be moved to the Miscellaneous category. The user should then move them to the appropriate weight/height based color group.

Protocol Management continued

- It is not possible to select a Default Protocol for the following Pediatric protocol categories: Neck, Upper Extremity, Chest, Abdomen, Spine, Lower Extremity.
- If the weight of a child is at the cross over point of a weight category, due to rounding errors the correct weight category might not be selected. Please check the weight based category selected against the label and the patient's weight.

SMARTPREP – PURCHASED OPTION

- The film composers will flash and cannot be hidden on the SmartPrep display. This will occur if the Auto Film or Manual Film composer buttons are selected after confirm has been selected for SmartPrep. Don't select the film composer buttons while the SmartPrep display is first coming up or is visible to avoid this.
- If reconstruction fails during SmartPrep, scanning will pause and post a message. Press |Monitor Phase| to resume scanning for the monitor phase.
- The baseline image for SmartPrep may fail to display even though it was reconstructed and in the Browser. To continue, return to the ViewEdit screen and rescan the baseline image.
- The Monitor Phase image and graph information will not update if the table height is changed between the Baseline and Monitor phases. Do not change the table height after the base line scan has been acquired.
- If the Review Layout is the XR AutoView layout, after transition from Monitor to Scan Phase, the upper left viewport (XR) will be blank. Move the mouse cursor to the XR viewport to redisplay the image.
- If Monitor Phase image is roamed or zoomed, the magnification factor and roam position will be carried through to the Scan Phase images. Check images as they begin to reconstruct for the Scan Phase to make sure they are displayed as desired.

PATIENT SCHEDULE

- Close the preferences screen before switching between Patient Schedule and New Patient.
- The Patient Name, Patient ID, Accession Number and Requested Procedure ID can only be edited if "Allow to Edit MWL" is set to yes in the Preference window.
- The Patient Schedule button may not display the Work List from the HIS/RIS server if the network is slow. Try again.

CONNECTPRO (PURCHASED OPTION)

- Bar Codes will fail to be read by the Bar Code reader if the HIS/RIS system the bar code was created on has a different language keyboard than the CT system. For example, if your CT system has a French language keyboard then your HIS/RIS must have a French language keyboard. If it is not possible to have the same language keyboard on each system, then manually enter the Accession or Patient ID number or select the desired patient from the Patient Schedule list to display the patient information on the New Patient screen.

ConnectPro continued

- Patients selected from Modality Work List will be displayed with ^ (carets) to define different DICOM fields. The carets are not displayed on the images.

PERFORMED PROCEDURE STEP (PART OF CONNECTPRO OPTION)

- PPS will post the message “Failed to Start” if the remote server is down.
- Scouts and Screen Save images created in ExamRx or in the Viewer on the Image Works desktop are not PPS aware in this release of software. Also, images created in Reformat, 3D, and Navigator are not PPS aware. For these image types, “INPR” will be posted in the PPS column in the Browser even though PPS is not enabled.

RECONSTRUCTION

- If images fail to recon the following should be used in order:
 - Select [Recon Management], [Unsuspend Queue]
 - If the image still fails to recon
 - Select [Recon Managment], [Restart Queue]
 - If the image still fails to recon
 - Select [Service Desktop], [System Resets], [Recon Reset]
 - If this still fails
 - Shutdown and reboot the system.
- Lung Algorithm
 - Provides edge enhancement between structures with large density differences, such as calcium and air.
 - Enhances the contrast of small objects. For best viewing and film quality, select a window width of 1000 to 1500 and a window level of -500 to -600.
 - Increases CT number values at the edge of high contrast objects. When planning to take CT number measurements of vessels or nodules in the lung, please check and compare your results with Standard algorithm images. (ROI and Histogram functions use CT numbers.)
 - The edge enhancement provided by Lung Algorithm may not be appropriate in some clinical cases. Please take individual viewing preferences into account when you choose Lung algorithm.
- Reconstruction for Cine acquisitions may occasionally suspend. Select Recon Management and unsuspend suspended queue entries.
- Recon retries may be seen for 1i cine acquisitions.
- Images at a 50 DFOV may show a shaded band at the edge for the DFOV. Change the DFOV to 49.8 to avoid the shaded ring.
- Images of anatomy acquired at the edge of the SFOV may be smoother than if the anatomy had been scanned at iso-center. Take care with positioning extremities which are being imaged at the side of the patient such elbow, wrist, shoulder. Move the side of interest as close to iso-center as possible.

Reconstruction continued

- Retro recon may not be able to get the same image locations as prospective recon. This is due to rounding in the start and end locations. To avoid this mismatch, prescribe the start and end locations in even numbers.
- Retro Recon lists Service exams from Fast Cal. Do not attempt to recon this data, the SRU will shutdown. Service exams are listed with exam numbers that begin with 50000.
- SmartStep series are listed in the Retro Recon exam list but cannot be selected.

RETRO RECON

- If a 1 rotation 4i or 2i axial scan is prescribed using add group in ViewEdit with a Superior to Inferior scan direction is retro reconned, the resulting images will be reconned Inferior to Superior. The image locations are correct, but the image numbers will not match the prospective images.
- If you delete queued Retro images the image space reserved for those retro recons is not given back. You will not get the space back until you reboot the system.
- Only start Retro Recon when scanning is complete. Do not delete queued retros while scanning is active, scanning could stop.
- If maximum A-P or R-L offset is selected in Retro Recon, one image may fail to reconstruct. Select an offset that is 0.5mm less than the maximum value allowed.
- The start/end locations of a retro reconstruction may be different from Recon 1 and 2 and 3 when a PMR with a thicker slice than Recon 1 is prescribed for the series.
- Retro Recon may hang and core. If the Retro recon screen fails to display or queue retrospective images, retro recon has cored. Reboot the system to recover.
- Retro Recon displays 1.25 and 2.5 as selectable slice thickness for a (SmartSlice)ThinTwin sub mm acquisition. Only 0.63 slice thickness will be allowed.
- If a R, L or A, P value is more than half of the Display Field of View in mm. The image annotation will not annotate R and L or A and P on the images. For example if the DFOV is 10cm and the R value is 56mm then the image annotation will show R R. Use a R-L or A-P value which is less than half of the DFOV in mm to avoid this.
- It may be possible to confirm a retro recon prescription with an invalid recon start and end location that results in a recon suspend. To avoid this, change the image thickness before changing image locations.

RECON MANAGEMENT

- Scout queue entries can be selected in Delete Retro queue entries. Only select Scout queue entries to delete if you are sure you do not need the scout image reconstructed.
- Before saving or restoring scan data make sure the system is idle and no Archive, Network or Filming is active. No other features should be accessed until the save or restore is complete.
- Recon Management may hang while trying to display the menu. Finish the current exam if scanning and shutdown and restart the system to correct the problem.

Recon Management continued

- If you wish to cancel a Save or Restore scan data, the cancel button will cancel only after the current scan file is saved or restored. A helical scan file contains a large amount of data even though it is one scan file, that file can take up to 30 minutes or more to save. You are unable to scan patients while saving or restoring Scan Data.
Make sure that you have ample time to complete the save or restore before beginning.
- Scan data save or restore is active when the dialog message indicating saving or restoring scan file is displayed.
- If saving scan data, after the save is complete select Restore Scan Files to verify the data is stored.
- Scan File save is not complete when the Saved File message is displayed. The file is saved but the MOD is starting the unmount task. You may have to wait 5 to 7 minutes for the MOD to unmount. The write light on the MOD will turn green when the MOD is unmounted.
- Scan Files restored from a different system will be reconstructed using the product name of the system it is restored on. For example, restore scan files from a H1 SDAS QX/i or a H2 SDAS Plus to a H3 MDAS Ultra and the product name on the images will be indicated as LightSpeed Ultra instead of LightSpeed QX/i or LightSpeed Plus. All other patient and parameter information is correct.
- Scan data save is limited to the size of the MOD. Use 2.3GB MOD for saving scan files. The following table lists the maximum exposure time to avoid this problem. You can also verify in recon management the size of the scan file before attempting to save the scan file.

Rotation Time	Max 4 slice scan time	Max 8 slice scan time
0.5 sec	59.3 sec	29.6 sec
0.6 sec	59.6 sec	29.8 sec
0.7 sec	59.6 sec	29.8 sec
0.8 sec	67.8 sec	33.9 sec
0.9 sec	76.6 sec.	38.3 sec
1.0 sec	84.8 sec	42.4 sec

VARIVIEWER

- If the scout is taken after the series used for VariViewer operation, the system may be unable to find the scout for graphic batch prescription. Manually enter the range for the batch prescription.
- The DFOV for a VariViewer session may be incorrect. The DFOV annotated on the images is the prescribe DFOV but visually it can be seen is incorrect. Any measurements done on these images will be incorrect. Select Batch mode and set up a batch prescription. This forces the system to perform a DFOV calculation and then the images will be displayed with the correct DFOV.
- If you resize the viewport in VariViewer, the image may not initially be displayed. Use the Page Turner in the lower left corner of the viewport to toggle to the time.
- Switching back from a 768 viewport in VariViewer causes the VariViewer to be displayed forward instead of the AutoView viewport. Use the pager turner to toggle back to the AutoView viewport.
- VariViewer session may not be able to be started if there multiple Pause scans during the acquisition of data.
- VariViewer is not compatible with a gantry tilt.
- It is possible to get two different DFOV for a combined VariViewer session if Column Edit is used. Use Row Edit for combined session with VariViewer.
- Any zoom applied to a VariViewer image in not maintained as you page through the images.

SMARTSTEP – PURCHASED OPTION

- The ExamRx Display process may crash if you confirm a SmartStep series from a desktop other than ExamRx Display. To avoid a possible crash, always confirm SmartStep from the ExamRx Display desktop.
- Retro Recon is not available for SmartStep acquisitions.
- The X-ray On message is not seen if you are doing a list select at the same time an exposure is being made.
- Screen Saved images created during a SmartStep session may fail to show up initially in the List Select Browser. The images will appear in the Browser after exiting SmartStep.

DIRECT3D – PURCHASED OPTION

- Some error messages for Direct3D may refer to the feature as VAV. Initially, the feature was called Volume AutoView.
- Direct3D is not compatible with Biopsy, a tilted gantry or with multi-slice prospective reconstructions.
- If an invalid number for Zoom is entered, the system defaults the entry for the field to 0.1.
- When creating Screen Saves, wait for the image to re-render completely before pressing [Screen Save] or only the portion of the volume which was rendered will be captured in the Screen Save.
- If Standard resolution is used for data sets with 256 or more images, the rendering will have corrupt data. Use Enhanced for large volumes (250-500 images).
- Direct3D rendering may have erroneous data at the edge of the volume for standard resolution renderings when clipping is used. To eliminate this data, turn clipping off.
- Advantage Windows 4.0 must be running version 4.1 software to support the Direct3D Save State Series.
- Upper Extremity curve does not display a usable 3D volume rendered model. Please select Lower Extremity curve for musculoskeletal models.
- You may see the border of the Direct3D viewport turn red when review layouts are switched.
- The window level for the Direct3D viewport is applied to all viewports when review layouts are switched.
- The Direct3D model may not initially build properly. Enter Interactive Review and re-render the model and then the model will build correctly.
- If you resize the viewport in Direct3D, the image may not initially be displayed. User the Page Turner in the lower left corner of the viewport to toggle to the time.

CT PERFUSION 1 ON OC – PURCHASED OPTION

- Saving TIFF files of the graph and images is not supported because there is no floppy drive hardware available on the OC.
- The CT Perfusion Control Panel may be displayed on the current desktop if CT Perfusion is launched from the Browser and the desktop is changed before CT Perfusion screens are completely displayed.

CT PERFUSION 2 ON OC – PURCHASED OPTION

- Saving TIFF files of the graph and images is not supported because there is no floppy drive hardware available on the OC.
- Saving TIFF files of the graph and images is not supported because there is no floppy drive hardware available on the OC.
- Set the color ramp to gray scale for screen saves images if you are going to network the images to another station such as PACS to improve quality of the screen save.
- Initially as CT Perfusuion2 is loading, you may see garbage flash along the left side of the screen.

SMARTSCORE PRO – PURCHASED OPTION

- Only 0.8 and 0.5 second rotation speeds are valid for Prospective Gating.
- A backslash (/) in the Patient ID for SmartScore exams will cause the patient report to fail to print.
- If the Confirm button is not available, select the Gating button and turn gating off and click on Accept. Select Gating button, again, and turn gating on and click on Accept. The Confirm button should now be available. Contact your local service representative if you experience this issue for further investigation.

CARDIAC SNAPSHOT – PURCHASED OPTION

- If a lead falls off during acquisition, the heart rate annotation will be inaccurate for the portion of the acquisition after the lead falls off. Images reconstructed for the complete acquisition using Snapshot Segment will initially display correct BPM for the period of time the leads were connected, and then ungated images will be displayed followed by images annotated with abnormally high heart rate where the leads have fallen off.
- If the Gating button is displayed in red and heart rate is seen, this is most likely due to loose connection of the cable at the connection to the EKG monitor or the scanner. Make sure you see a heart rate before you begin the scout scan.
- CardIQ Snapshot Segment annotation is not displayed on the images displayed on the Advantage Windows workstation. The average heart rate and the percent of R-R interval will not be seen. The Cardiac Helical scan type will be annotated as Axial.

Cardiac Snapshot continued

- Snapshot Segment images can be networked to a PACS system; however due to an error in the DICOM header, they can be retrieved from the GE PathSpeed PACS system but the image annotation is wrong.
- Annotation for Snapshot Burst and Burst Plus may not reflect the number of sectors (SBB2, SBB3, SBB4) that were used to create the images. To verify the number of sectors used to create the images check the temporal resolution annotated on the image.
- Temporal Resolution for a Segment image for a single sector is 0.33sec (330ms) for 0.5 second rotation and 0.4sec (400ms). If the rotation time is 0.5sec, the temporal resolution for a 2-sector image will be between 0.33second and 0.17second (influenced by heart rate), a 4-sector image will be between 0.08second and 0.11second. If the rotation time is 0.6, the temporal resolution for a 2-sector image will be between 0.4second and 0.2 second (influenced by heart rate); a 4-sector image will be between 0.1second and 0.13second.

AUTO TRANSFER

- A series may fail to Auto Transfer if [Next Series] then [Create New Series] is selected after a scout scan is prescribed, but not scanned, i.e.; pause and return to view edit is selected.

DISPLAY

- The Exam Rx List Select and Image Works browsers may in some cases not list an exam, series or image. This may be seen after an interruption of power to the system or the database may be updating slowly. The patient does not need to be rescanned.
- Verify the following:
 - Type the Exam, Series, Image on the accelerator line to display the image,
 - Select another exam in the browser this will cause the browser to update,
 - The browser may update after a slight delay,
 - Reboot the system,
 - A second reboot may be needed if a power fail has occurred.
- The browser may update slowly to reflect Screen Save images created in an exam. If the Screen Save images are not listed in the browser wait for them to appear. The time to appear may be up to 15 minutes if the Automatic Database check and recovery is in progress when the Screen Save images are created.
- In general, wait for a display action to complete before entering in another command.
- With the system configured in French or German language, if a comma is entered in explicit magnify, i.e. 2,3 the comma will not be interpreted as a decimal. The magnification factor is applied without the decimal. In the example shown, 2,3 the image would only be magnified 2X not 2.3X. Always enter a period to designate a decimal and have the correct magnification applied.
- Cross Reference lines will post incorrectly if sort by image location is selected. Make sure that sort by number is selected before posting Cross Reference lines.

Display continued

- Cross Reference lines may not post with some combinations of DFOV and RAS center. RAS centers at the edge of the DFOV and small DFOV's will exhibit the problem. Retro reconstruct the images to a larger DFOV if posting of the cutlines is desired.
- Image annotation may overlap or not be fully shown if 32 character patient name is entered for 3D, Navigator, Reformat, MID formats, DentaScan images.
- Text and Series pages only display 24 characters for Patient Name.
- Text pages will not show foreign characters entered using the ALT GR key.
- ROI and STD deviation numbers may be reported as zero after a Zoom is applied to an image with a ROI posted. This is due to partial pixel contained in the ROI. Adjust the size of the ROI to include full pixels and recalculate the ROI.
- No text will be entered in the accelerator line if the the mouse cursor is over the film composer.
- Some combinations of E/S/I will fail to display an image, if cap locks is on. To select an exam series and image from the accelerator line use the following format when cap locks is on: For example to select Exam 576 Series 2 image 3 type E576 2 3 on the accelerator line.
- Selection of an exam series or image may fail a second time after a exam series or image is displayed. If the desired exam series or image does not display then select the desired images from the browser.
- Image location seen on the scanner may differ from that seen on an AW 3.1 system for tilted images. LightSpeed reports the image location as iso-center of the image. AW 3.1 reports image location as the image center for the image. With tilted images there will be a difference in the location numbers proportional to the off center distance times the sin of the tilt angle. The image location can be reported as image center on LightSpeed by using the Report Cursor function.

EXAM RX DISPLAY

- Display may reset it's self after a software problem. You may see display stop or not accept your mouse input. Wait for a few seconds and display will automatically reset. You will have to redisplay the images you wish to work on.
- The MIROI pop up may display but will not function. This will occur if a Auto View or Cross reference viewport is in primary focus (blue border) MIROI is not valid in these viewport types. Place a Free or Auto Link viewport in primary focus and reselect the MIROI button to continue.
- Any Report Pixel or MIROI chart which is not screen saved, but is filmed will show the Exam Series and Image as 1000/1/1 on the film composer icon if E/S/I is selected. This is the exam/series/image number the system assigns to this chart display The patients exam number is listed correctly on the filmed image.
- Image selection from the accelerator line will not function if the primary viewport contains a MIROI plot or Report Pixels chart. Use List Select to display a new image.
- Any accelerator commands entered for Series Binding, Annotation levels for display or filming will be saved as defaults.
- ExamRx Display may hang. In some cases switching to the Image Works desktop and back to Exam Rx display may clear the problem. If this doesn't work then perform a system shutdown and reboot the system.

ExamRX Display continued

- The right mouse may fail to roam even when the state is set to roam. A system shutdown may be required to clear this problem.
- Display may crash when selecting a 3D object from the List Select Browser. When a 3D object is selected, 3D should be selected from the Browser menu.
- Images may not be displayed after a switch between Auto View or Image Review layouts.. This is due to a restart of the display process. Redisplay the desired images in viewport to continue.
- The first time the trackball is used for paging after a system reboot, window level for the image will be changed even though the P icon is displayed in the lower right corner of the viewport. Click once to switch to W, then click again to
- The mouse may become locked in either the roam or zoom mode in the AutoLink viewport. Even though the other operation is selected the mouse will remain in the roam or zoom mode. Change layouts first, if that does not release the mouse from the roam or zoom mode; then reboot the system.
- Viewport format may not be able to changed and requires switching layout to correct. This occurs if F3 or Shift F3 is used for filming and next image is used. These steps lock the viewport. You will have to switch to another Review layout or AutoView layout to correct.
- Drag and drop for a scout with a large number of cross reference lines may take longer to capture. Click left and hold the mouse on the scout with cross reference lines for a second before dragging to film composer.

IMAGE WORKS DISPLAY

- WW and WL adjustments will only be maintained on individually selected images if the middle mouse button is used. If the accelerator line or Presets are used all images will update regardless of primary or secondary focus of the images.
- Any accelerator commands entered for Series Binding, Annotation levels for display or filming will be applied and not held if the viewer is closed. Use User Preferences in the viewer to save the settings as a default if desired.
- Window/Level does not display stored value set in the Viewer when +/- series is used. The W/L used is the same as the last series displayed.
- Images created in Add/Sub are displayed with value of WW 4098 and WL 1024.
- If paging is selected while the system is in Compare Mode, the upper left viewport is left blank. Return to the Browser and select the desired series for paging and restart the Viewer.
- Accelerator Command Line entries for E/S/I do not act properly. The first image in the series is always displayed instead of the requested image within the series.
- Cross Reference line numbering along the top of a Sagittal scout may be listed in offset manor.
- Square Viewport selection to ON in User Preferences is not saved after a reboot. If Square Viewports are desired, they will need to be selected each time the system is rebooted.

EDIT PATIENT DATA

- The system may hang if Edit Patient is started while the following operations are in progress:

Network Receive

Prospective or Retrospective reconstruction is active

Archive Restore

To avoid this, confirm Edit Patient during idle times in network receive, active recon or archive restore.

- Patient age may be seen as zero in Edit Patient Data even though a birthdate is entered. Enter the birthdate again and patient age will be calculated.
- If duplicate accession numbers exist, Edit Patient Data will not allow the exam to be modified.
- If there is a space at beginning or end of the hospital, Edit Patient Data will fail.

3D/NAVIGATOR/DENTASCAN/ADD/SUBTRACT

- These applications can only load 1000 images. Highlight the desired range of images by selecting the first image, then hold the SHIFT key down and highlight the last image desired for the application.

ADD/SUBTRACT

- Comb images will not contain e/s/i information used to create the image.

REFORMAT

- When selecting reformat wait until the % loading dialog is displayed before switching to the Exam RX desktop. This will avoid having to select Load slices to start the reformat model build when returning back to the Image Works desktop.
- The reformat location will not be annotated on images filmed when Printer is selected in Batch Mode. If it is desired to have the reformat location annotated, then use Save in Batch Mode and film the images through Print Series.
- In Batch mode, images reconstructed using "Rotation" or "Oblique" modes will start with number 2 while images created using "First View/Last View" start with 1. In Rotation and Oblique modes, a plan of the reconstructions is screen saved and becomes image 1 for the series.
- Occasionally, the color for the viewports and control panel will change because there are not enough colors available in the resource file to support reformat. To recover, quit reformat and restart the application.
- Window/Level for images are not displayed using the value stored in the list. Use the window/level function keys or select within Reformat to reset the W/L.
- Reformat will only allow filming in Batch to the first 30 cameras in the list. If another camera is desired save the batch images to disk and use Print Series to film the images.

Reformat continued

- Reformat has trouble loading images with different parameters. Normally reformat will only load the images that have similar parameters even if the series contains groups with different parameters. To avoid this, selectively highlight the images for the desired group.
- The maximum number of images for Reformat Detail is 1000 images. If there are more than 1000 images use Reformat Standard or limit the number of images to 1000.

FILMING

- Images or Auto Film control buttons may fail to display in the AutoFilm viewport. This may occur after a switch between desktops. To display the images change to a different AutoView layout, then switch back to your desired layout.
- Keep current on Auto Film.
- Any of the Auto Film control buttons may be activated when the cursor is over the button and the space bar is selected. Place the cursor in the Auto Film viewport only when needed to avoid this problem.
- Auto Film may fail to film. Auto Film status shows paused. Toggle the [Pause] button in the Auto Film viewport and select [Start New Sheet] or [Continue Same Sheet] to restart filming.
- Occasionally, the last image will not print. Cancel Auto Film and manually film the image to the Auto Film Composer. If scouts and reference lines were prescribed, they will need to be filmed manually.
- Auto Film may fail to display images if recon is having trouble reconstructing the image or if images have failed to install in the data base. This may also occur if a large exam has not reconstructed all images and scanning is started on a new exam. The following dialog will be posted if Auto Film cannot display images:
 - Exam 100
 - Series 2
 - Cannot find 10 images
 - Skip Missing images Continue Cancel Film Series

You are presented with 3 choices:

1. Skip Missing images will skip the images which Auto Film cannot find. For example if Auto Film had filmed images 1-10 and you then got the message that 10 images were missing and selected Skip Missing Images, Auto Film would then start filming again with image 21.
2. Continue will look for the images again, if they are not found then the dialog will be posted again. Before selecting continue verify that recon is active, that the images have been reconstructed, the missing images are not suspended or paused in the Recon Queue and the images are able to be displayed. If the images are suspended or paused in the recon queue, then unsuspend or restart recon to reconstruct the images before selecting Continue.
3. Cancel Film Series will cancel Auto Film for the series currently being Auto Filmed. The images will then need to be manually filmed.

If you get this dialog because you have begun an exam while the previous exam still has images to reconstruct, then Select Continue and select Pause Auto Film, resume Auto Film when the Exam Series and Images from the previous exam have reconstructed.

Filming continued

- When prescribing Mag factor for auto film, scout images if the scout is longer than 500 mm then use a magnification factor less than 1 to display the entire scout.
- The manual film composer may display when confirm is selected for scanning if the composer had been closed by selecting iconify in the upper right corner of the composer.
- It's best to run Print Series from only one desktop at a time.
- It's best to resolve any paused queue entry as soon as possible.
- Print Series may pause automatically under heavy system load.
- If Anonymous Patient is selected for an exam where Auto Film is in progress, the system may fail to cross reference the slices on the scout. If Anonymous Patient is required for series which are still active in Auto Film, wait till Auto Film is complete for the series before proceeding with Anonymous Patient.
- To install a SMPTE pattern select the following. Once installed the images will be available for viewing from the Image Works browser or List Select in Exam RX. And will be listed as Exam 1000. The patient Name will be listed as SMPTE.
 - Select Service Desktop
 - Select [Image Quality]
 - Select [Install SMPTE Image]
- The format built in a protocol may be changed when the protocol is used with a message that the format was changed due to an invalid format, even though the format is valid. Verify in Auto Film set up is the format you desire before confirming scan.
- For Multi-Image Display (MID) in AutoFilm, filters, gray scale and orientation selections are applied only to the image in the upper left viewport. If there are an odd number of images in the series, the filters, gray scale and image orientation is applied to the first pass for filming.
- If the F1 key is used to film the Text Page ROI, Exam Text Page and Series Text Page, the text pages will be filmed to the film composer last selected in the pop-up for text page. Use the filming selections in the text page screens to film the information displayed.
- The message "Film Formatting in progress. Please retry." may be seen when filming to the Manual Film composer when the system is busy with other simultaneous functions such as recon, network, autofilm and scan. Retry when the system is idle. A system reboot may be needed to recover.

ARCHIVE

- The feature status area or the browser may report an Exam has saved even though all images have not been saved to MOD. Scroll through the images in the image window. Highlight the images listed as Archive N and resave them.
- Restoring Exam, Series or images that already exist on the system disk will not post a message that the images are restored or that they already exist. If you have restored images, but get no message that they have been restored, verify that they don't already exist on the system disk.
- When the archive media gets close to being full, the system will always look to see if the exam, series or image can fit on the MOD, if it is desired to have exams sequentially on a MOD, place a new MOD in the drive when first notified the disk may be full.
- If Label is selected in Archive, finish the Label process, don't select cancel on the Label pop up. Archive will be non functional, a system reboot will be necessary to recover.

Archive continued

- It's best to resolve any paused queue entry as soon as possible.
- To minimize corruption of MODs, it is extremely important the MOD media be detached and removed from the MOD drive before doing a shutdown or recycling power to the system.
- When restoring from a MOD, it is recommended to write protect the media before placing it in the drive. If the system is unable to read the media, remove the write protect and see if the system is able to preform a recovery of the media to access the data.
- If the MOD can not be detached or will not eject from the drive, do a shutdown and eject the MOD once the message "Hit any key to start the system" is seen. Do not use the screw to eject the disk.
- If you get a timeout message when trying to access the MOD, the MOD drive may no longer recognized by the system. Perform a shutdown to re-establish communication between the system and the MOD drive.
- Even though the Sony MOD drive states that it is a 5.3 gb drive. The archive software only supports a maximum size of 2.3 gb. 2.3 gb 512 bytes per sector is the preferred archive media size to utilize on the system.

NETWORK

- LightSpeed images will **not** transfer if Advantage Net protocol is used. Always use DICOM protocol to send LightSpeed images
- LightSpeed images **cannot** be sent to a HiLight Advantage, HiSpeed Advantage or CT Independent console. These systems do not support DICOM receive.
- Some 3rd party workstations may fail to receive LightSpeed scout images. This is due to the matrix size of the scout image. These stations do not support receiving matrix sizes greater than 512. Some LightSpeed scouts images have matrix sizes greater than 512. If it is desired to have the scout image on the workstation, Screen Save the scout and then transfer the screen save image to the workstation.
- If you query a LightSpeed system from a CT/I or Advantage Windows system in Advantage Net, only Advantage format image exams will be displayed, No LightSpeed exams will be shown. Always query a LightSpeed system using DICOM protocol.
- Lateral scouts displayed on an AW 3.1 system will initially display with zero rotation; they should display with a 270-degree rotation. Rotate the image in the left direction using the rl command on the command line to display the scout in the desired format.
- Images networked to a Advantage Windows 3.1 workstation running software versions prior to 3.1_07 will display a DFOV less than what is displayed on the scanner. This is due to the AW not taking in account the pixels under the focus border of the viewport.

Network continued

- Images networked to a AW 3.1 will have the following annotation missing:
 - Accession number,
 - Date of Birth,
 - 4i, 2i or 1i recon mode on Axial, Retro Axial or Axial Reformatted images,
 - Series type on Retro images,
 - Table Speed on Helical images,
 - Images are annotated +C when Oral Contrast is used. View the Series Text Page to see if IV contrast was used.
 - CardIQ annotation for BPM and percent of R-R interval and scan type will displayed as Axial.
- The number of images indicated in a series may be incorrect on the remote browser when a query is made from a AW or CT/I station.
- Series types may be listed differently when images are networked to an Advantage Windows 1.2, 2.0 or 3.1 system than that shown on the LightSpeed system
- Exams with Swedish, German or French characters in any Patient Info field will not be transferred to a Advantage Windows.
- When using DICOM protocol, the entered host name must match in spelling and case sensitive, otherwise a connection error message will be displayed.
- It's best to resolve any paused queue entry as soon as possible.
- If mobile software is installed on the system, the scanners internal subnet (192.9.200) should be used for an AW system or network printer so the IP addresses will be the same for each site. However, some cameras may require access to the external hostname for the AE identifier.
- When networking from a remote host and it is shutdown before the transfer is complete, you may receive a corrupted image. The corrupt image should be removed and retransmitted when the remote host is up again.

IMAGE MANAGEMENT

- Don't let image space fall below 200 images on a single disk system (20,000-image storage). This will ensure there is room on the system disk to confirm scans and to install reconstructed images.
- Removal of images should be done with the system in an idle state. Failing to remove images at idle times can cause image install timeouts that can then cause recon to shutdown and can cause the browser to fail to respond. The user may also see a blank list in the browser. The system should not be shutdown until the image space has been updated fully from the remove. The length of time to synchronize the database is proportional to the number of images being removed. This will assure that all images are reconstructed, displayed and installed into the database without error and that image space is update correctly.
- It may not be possible to remove all series for an exam that has a SmartStep series in it. The exam will be able to be removed completely after a system reboot.

MOBILE

- Mobile systems scanning using the back-up power generator instead of shore power will experience vibrations as the generator compensates for sudden electrical loading as the X-Ray turns on. This is normal operation for generator verses shore power. To ensure these vibrations experienced on generator power do not affect image quality, the anatomy being scanned should be secured to the cradle using the restraint straps.
- The CT Mobile software allows the user to pre-program different site names and networking addresses for sites visited by the mobile. If the system configuration is moved to an external subnet instead of the internal subnet, the network for the camera and AW will need to be redefined for the external subnet.

Page Intentionally Left Blank

g

GE Medical Systems

P.O. BOX 414

MILWAUKEE WISCONSIN 53201-0414 U.S.A.